CHECKPOINT: ARCHITECTURE AND DESIGN II
This checkpoint is all about observing a video. So after watching the video, I will discuss the purpose, tools used and how to manage centralized authentication.

WHAT'S THE PURPOSE

The purpose of configuring centralized authentication using RADIUS, Active Directory, and network devices (like UTM1) is to make sure they simplify, secure, and centralized user authentication across an organization.

Centralized User Management

Unified Access Control: All user credentials are stored and managed in one location (Active Directory), making it easier to administer permissions and maintain user accounts.

Streamlined Administration: Adding, updating, or removing user accounts in AD automatically applies to all connected RADIUS clients (e.g., UTM1, VPNs, or wireless access points).

Scalable Integration

Device Compatibility: RADIUS is widely supported by various network devices (e.g., firewalls, UTM appliances, VPN servers, and Wi-Fi access points), ensuring a unified authentication experience.

Centralized Logging and Monitoring

Activity Auditing: Logs generated by the RADIUS server (NPS) provide detailed information about authentication attempts, allowing you to track user activity and detect suspicious behavior.

Compliance and Accountability

Regulatory Compliance: Centralized authentication supports compliance with regulations (e.g., GDPR, HIPAA, or ISO 27001) by maintaining secure, auditable user access records.

Accountability: RADIUS logs link access attempts to specific users, providing traceability for security incidents.

Use Case Examples

- Corporate Networks: A business with employees using VPNs, Wi-Fi, or remote desktops can secure access via centralized authentication.
- Education Institutions: A school or university can manage staff and student access to network resources through one directory.
- Small to Medium Enterprises: SMBs can integrate their UTM devices with AD to manage firewall, VPN, and web access with minimal overhead.

- By implementing centralized authentication with RADIUS and Active Directory, organizations enhance both security and efficiency while simplifying the management of user access.

Tools Used in the Centralized Authentication Setup

- Windows Server
- Active Directory (AD)
- Network Policy Server
- UTM1 (Unified Threat Management Appliance)
- VMware
- Event Viewer
- Workstations or Test VMs
- Networking Tools e.g Wireshark, Ping/Traceroute, RADIUS Test Clients

These tools work together to provide a robust, secure, and manageable centralized authentication system for your lab or production environment.